

Movie Ticket Booking System

Functional Requirements

Requirement	Description
Movie Listing	The system shall display all movies currently playing in the cinema, including their title, language, and duration .
Show Listing	For a selected movie, the system shall display its available shows with the screen number and start time .
Seat Availability	For a selected show, the system shall display the seat layout with each seat marked as AVAILABLE or BOOKED .
Booking	A customer selects one or more seat numbers for a show. If any selected seat is already BOOKED , the whole booking is rejected and no seat changes state. Booking is confirmed only after payment succeeds .
Pricing	The system shall calculate the booking amount based on seat type: SILVER ₹150, GOLD ₹250, and PLATINUM ₹400 .
Payment	Exactly one method (UPI / Card / Cash) shall be used per booking. If payment fails, seats are released and booking status becomes FAILED .
Ticket Printing	After successful payment, the system shall print a ticket containing the booking ID, movie, screen, time, seat numbers, and total amount .
Cancellation	A customer shall be able to cancel a booking using the booking ID . After cancellation, the seats associated with that booking shall become AVAILABLE again.

Non-Functional Requirements

Requirement	Description
Modularity	The system shall be modular, with one class per file and each class having a clear responsibility.
Extensibility	Adding a new payment method, such as NetBanking , shall require adding a new class without modifying the existing payment classes or BookingService.
Input Validation	The system shall validate invalid seat numbers and menu choices , display a clear error message, and continue execution without crashing.
Maintainability	The system shall use clear names, small functions, and constants so that the code is easy to understand and modify.